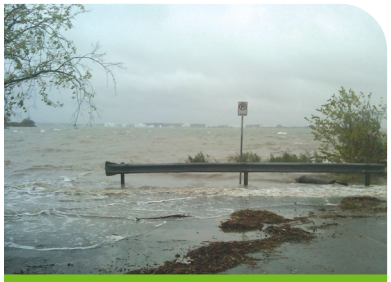
RESILIENT NORFOLK

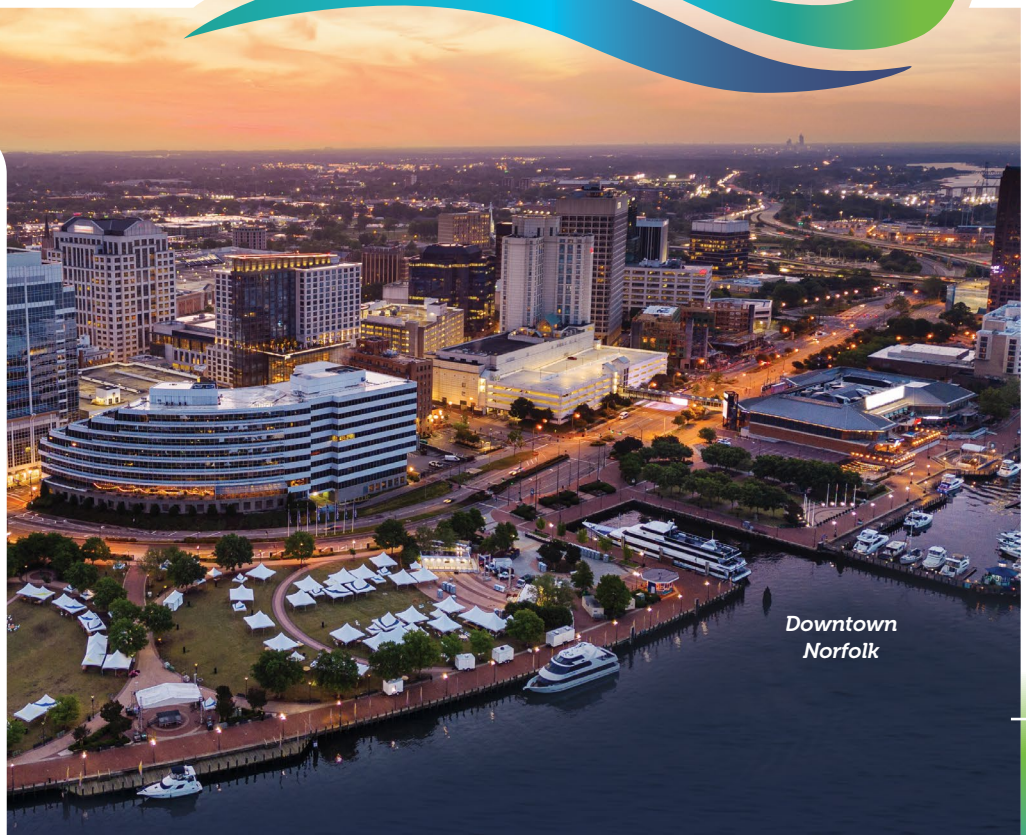
COASTAL STORM RISK MANAGEMENT

3rd Bay and Pretty Lake Avenues

15th View Street



48th Street Dead End



Downtown Norfolk

OVERVIEW

The City of Norfolk is increasingly at risk of flooding and damage from coastal storms. Norfolk has one of the highest rates of relative sea level rise (RSLR) among Atlantic coastal communities. The Coastal Storm Risk Management (CSRM) Project will reduce the City's risk from coastal flooding and damage from nor'easters, hurricanes, and other significant storm events.

PARTNERSHIP

The Norfolk Coastal Storm Risk Management Project is a collaboration between the USACE and the City of Norfolk.

THE CITY OF NORFOLK
OFFICE OF RESILIENCE

US Army Corps of Engineers®

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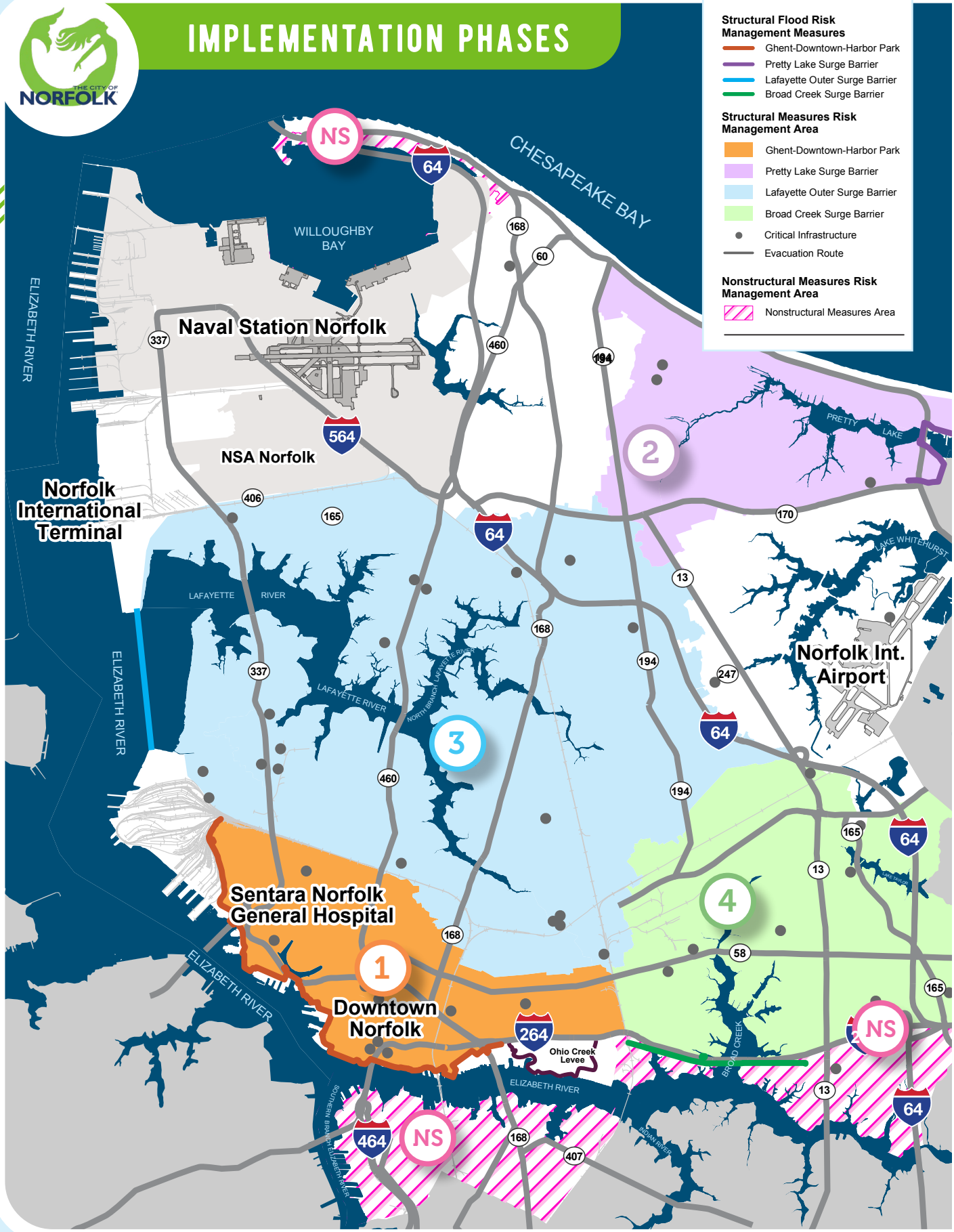
IMPLEMENTATION PHASES



PROJECT FEATURES

The CSRM project includes structural, property-specific, and nature-based features.

Structural Projects: Structural mitigation projects are construction projects intended to lessen the impact of a coastal storm. Examples include floodwalls and levees, storm surge barriers, and tide gates.

Property-specific, nonstructural: These measures involve the modification of existing buildings and structures. Examples include property floodproofing through FEMA grants, CSRM project nonstructural adaptations such as basement fill, and protecting critical facilities such as a hospital.

Nature-based Solutions: Nature-based solutions preserve or restore natural areas and their protective functions. Natural areas could include floodplains, wetlands, barrier islands, and sand dunes. Examples include beach and dune preservation, tree planting and preservation, and wetland preservation and restoration.

Neighborhoods on Norfolk's southside have requested re-evaluation of the features proposed for their area. The re-evaluation process is underway with elevation of the request to the federal government.

THE CITYWIDE PROJECT IS DIVIDED INTO 5 IMPLEMENTATION PHASES

1 Downtown

A system of floodwalls with a levee, surge barriers, and nature-based features, extending from Ghent through downtown connecting to the Ohio Creek Watershed project.

2 Pretty Lake

A system of floodwalls and storm surge barriers to reduce storm surge from entering Pretty Lake at Shore Drive.

3 Lafayette

A storm surge barrier from Norfolk International Terminal (NIT) to the Lambert's Point area to reduce storm surge risk to the Lafayette River watershed.

4 Broad Creek

A system of floodwalls, storm surge barriers and tide gates to reduce storm surge from entering Broad Creek at I-264.

NS Nonstructural

A series of property-specific flood mitigation projects: home elevations, basement fills, floodproofing, etc. Work would take place in multiple areas of the city including Berkley, Campostella, Elizabeth Park, Ingleside, Willoughby Spit, Woodbine, and River Forest Shores.

RISK REDUCTION FEATURES

The CSRM project will utilize several types of flood improvement structures that are specifically designed for reducing coastal flood impacts.

T-wall - A sturdy concrete wall, shaped like an upside-down T. It consists of a concrete base with tall floodwall panels extending upward.

Levee / T-wall - This combination is a T-wall with compacted, heightened material on one side to provide additional support.

Bin-wall - A gravity retaining wall system made from adjoining closed-faced bins, then backfilled with soil.

Levee - Man-made structures, such as an earthen embankment, designed and constructed practices to contain, control, or divert the flow of water.

Property-specific, nonstructural - These focus on reducing the damages caused by flooding to homes and businesses.

Critical Infrastructure - These flood mitigation solutions will protect the assets, systems and networks that are essential to the City's function.

Nature-based features - These are flood mitigation solutions that mimic natural processes and include varied practices that can be applied at many different scales.

Gate - Adjustable gates used to control water flow in flood barriers, reservoirs, rivers, streams, or levee systems.

Pump station - These help protect areas by pumping away large volumes of water, preventing the occurrence of flooding.

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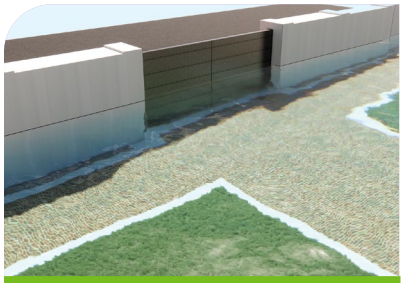
COASTAL STORM RISK MANAGEMENT

PROJECT AT A GLANCE

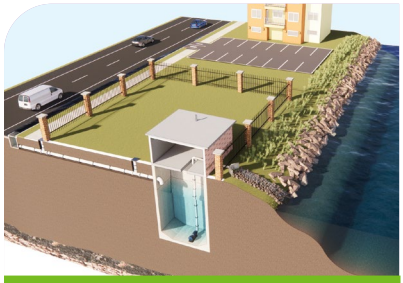
The $2.6 billion project features, storm-surge barriers, nearly nine miles of floodwalls and levees, 11 tide gates, and ten pump stations, along with a series of nonstructural projects that include home elevations, basement fills and commercial proofing, and oyster reefs and living shorelines.

BUILDING RESILIENCE

With one of the highest rates of relative sea level rise (RSLR), the City of Norfolk's increased risk of flooding and damage during a coastal storm proves a resilience strategy is critical to strengthen our coastal community for the future.



Tide Gate



Pump Station

Home Elevation

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FOR MORE INFORMATION CONTACT

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Information contained within this document is accurate as of 11/13/25 and is subject to change.

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